

MANTHAN PATEL

Flutter Developer | Senior Mobile & ROS Engineer — Robotics and Real-Time Systems

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PROFESSIONAL SUMMARY

Senior Flutter Developer with 5+ years of experience and 25+ production mobile applications shipped across enterprise, fintech, education, and robotics domains. Core developer of the Odigo advertising and guidance robot platform, supporting 100+ live robot deployments across 2 countries (India and the UAE), integrating ROS, ROS 2, WebSockets, Serial/UART, and WebRTC into real-time production systems. Experienced leading a 3-person engineering team and delivering clean, modular, offline-first Flutter architectures with a strong focus on performance profiling and production reliability.

CORE SKILLS

Flutter, Dart, Android, iOS, Cross-Platform Mobile Development, ROS, ROS 2, WebSockets, WebRTC, Serial/UART Communication, REST API Integration, Firebase, Riverpod, BLoC, Provider, GetX, Clean Architecture, Offline-First Design, CI/CD, Codemagic, Agile, Team Leadership

WORK EXPERIENCE

Flutter Developer — Kody Technolab Limited

January 2022 - Present (4.5+ years) | Ahmedabad, India

- Architected and delivered 15+ production Flutter applications across enterprise, fintech, communication, education, and robotics domains, growing total career output to 25+ shipped applications over 5 years.
- Developed the Flutter and Android control applications powering the Odigo advertising-robot platform, enabling 100+ live robot deployments across 9 commercial sites in India and Dubai (2 countries).
- Cut robot-operator communication latency to approximately 24ms round-trip by engineering real-time interfaces integrating ROS, WebSockets, Serial/UART, and WebRTC, sustaining 10Hz telemetry with under 2ms jitter in production.
- Extended the Odigo platform to power Etihad Rail's Fujairah passenger station deployment, the UAE's first AI robot installation at a national railway station, by adding a multilingual voice assistant and real-time indoor navigation.
- Led and mentored a team of 3 engineers, introducing code review, sprint planning, and release processes that supported 10+ production releases.

Technologies: Flutter, Dart, Firebase, ROS, WebSockets, WebRTC, Riverpod

Flutter Developer — Saubhagyam Web Pvt. Ltd.

August 2020 - January 2022 (1.5 years) | Ahmedabad, India

- Delivered 10+ cross-platform Flutter applications from MVP to release for B2B and B2C clients across robotics, fintech, and e-learning domains.
- Integrated REST APIs and Firebase backends, enabling real-time data synchronization across all 10+ delivered products.

Technologies: Flutter, Dart, Firebase, REST APIs

KEY PROJECTS

Odigo Advertising-Robot Platform

- Deployed the platform live across 9 commercial sites: 7 in India (Jio World Drive Bandra, Lulu Mall Lucknow, Lulu Mall Trivandrum, Pacific Mall, R City Mall, DLF Mall) and 2 in Dubai (including BurJuman).
- Built ROS and ROS 2 integration spanning 3 communication protocols (WebSockets, Serial/UART, WebRTC) and live telemetry / operator interfaces, achieving secure, low-latency production deployment.

Flutter, Java, ROS, ROS 2, WebSockets, Serial/UART, WebRTC

Odigo - Etihad Rail (Fujairah Station, UAE)

- Delivered the UAE's first AI robot deployment at a national railway station by extending the Odigo platform for Etihad Rail's Fujairah passenger station.
- Built a multilingual voice assistant with wake-on-tap interaction and live indoor map routing, guiding passengers from the robot's current position to a requested destination.

Flutter, Java, ROS, ROS 2, WebRTC, Voice/NLU (multilingual)

Satsang

- Built a spiritual learning platform delivering daily satsang across 4 content formats (scriptures, katha, audio streaming, and video streaming), plus monthly digital magazines.

Flutter, Firebase, REST API, Audio/Video Streaming

Bonno

- Developed a premium pet care marketplace connecting users with verified caregivers across 4 service categories: boarding, grooming, sitting, and training.

Flutter, Firebase, REST API, Google Maps, Real-Time Tracking

Karuna Care

- Built a live-streaming contribution platform enabling real-time animal welfare support, featuring 2 integrated monetization tools: an in-app wallet and a coins system.

Flutter, Firebase, Live Streaming, In-App Wallet and Coins

Mamgo

- Developed a secure social messaging and calling platform combining 3 real-time capabilities: messaging, voice/video calls, and push notifications.

Flutter, Real-time Messaging, Voice/Video Calls, Push Notifications

Crimechime

- Built a community-driven crime reporting platform featuring real-time heatmap visualization.

Flutter, Laravel, Google Maps API, Real-time Database

TakeProfit

- Developed a crypto analytics platform combining 2 core tools: real-time portfolio tracking and a trader toolkit.

Flutter, Laravel, Firebase Auth, Real-time APIs

CERO / iCarica / Wuerth EV App

- Built EV charging network applications across 3 markets (India, the MENA region, and enterprise fleets), integrating OCPP for dynamic charger-status updates.

Flutter, Firebase, OCPP, Fleet Management, Navigation

Calling Vault

- Developed an encrypted calling and messaging application combining 2 privacy safeguards, anonymous numbers and encrypted calling, built on the Telnyx WebRTC SDK.

Flutter, Telnyx WebRTC SDK, Laravel

TECHNICAL SKILLS

Core Engineering: Clean Architecture, Modular Codebases, Offline-First Design, Performance Profiling, Render Optimization, Concurrent Communication, Production Reliability, ROS Integration

Flutter Development: Advanced/Adaptive UI, In-App Purchases (Stripe, Razorpay), Audio/Video Streaming, Voice and Video Call SDKs (Telnyx, ZegoCloud, Twilio)

State Management: Riverpod, Provider, GetX, BLoC

Storage and Persistence: Hive, Sqflite, ObjectBox, Get Storage, Shared Preferences

APIs and Data: REST (Dio, HTTP), Repository and Model-Based Architecture, Local Caching, Postman

Firebase and Auth: Firebase Authentication, Cloud Functions, OAuth (Google, Apple, Facebook, Twitter, LinkedIn)

Maps and Location: Google Maps SDK, Live Location Tracking, Background Geolocation

Release and CI/CD: Codemagic Pipelines, App Store and Play Store Releases, Multi-Module Version Compatibility

EDUCATION

Bachelor of Computer Applications (BCA)

Gujarat University | 2014 - 2017 | CGPA: 6.5

LANGUAGES

English: Professional working proficiency

Hindi: Native / bilingual proficiency

Gujarati: Native / bilingual proficiency